Classification models

Streaming Churn

Table of Contents

1. Business Understanding

a. Business Use Cases

At our streaming service company, we are facing a significant challenge with customer churn, currently at a rate of 17.75%. This is much higher than other streaming services, which is mostly between 5 and 10% (Canal, 2023). This high churn rate not only impacts our revenue but also increases customer acquisition costs as we constantly need to replace lost subscribers. The marketing team has identified an opportunity to proactively retain customers by offering targeted discounts to those at risk of churning.

In this project, we aim to develop a machine learning model that can predict which customers are likely to churn within the next month. This will allow us to implement a targeted retention strategy: offering a 50% discount for three months to those customers identified as high risk for churn.

This approach is motivated by several factors:

1. The high cost of customer acquisition in the competitive streaming market.
2. The potential to increase customer lifetime value by extending subscriptions.
3. The need for a data-driven, personalized approach to customer retention.

Machine learning algorithms are particularly relevant in this context because they can analyze complex patterns in customer behavior that may not be apparent through traditional analysis. It is also a more efficient way to process large volumes of customer data.

b. Key Objectives

The key objectives:

- Develop a machine learning model that can predict customer churn with sufficient accuracy to make the discount strategy economically viable.
- Reduce the overall churn rate by at least 50% through targeted interventions.
- Implement a cost-effective retention strategy that results in a net positive financial impact compared to taking no action.

Stakeholders

- Marketing Team:
  - Requires a tool to identify at-risk customers for targeted campaigns.
  - Needs the ability to prioritize customers based on churn probability.
- Finance Department:
  - Requires the retention strategy to be cost-effective.
  - Needs clear ROI metrics for the discount program.
- Customer Service Team:
  - Requires actionable insights to tailor their interactions with at-risk customers.

- Product Development Team:
    - Needs insights into factors contributing to churn to inform product improvements.

The project will address the stakeholder requirements by developing an accurate machine learning model to identify high-risk customers for targeted discount offers, coupled with a comprehensive cost-benefit analysis to ensure the strategy's economic viability. It aims to provide valuable insights into key churn factors, informing both marketing strategies and product development. The solution is designed to be scalable with existing customer management systems, facilitating ongoing use and refinement. By leveraging machine learning algorithms, the project seeks to create a precise and cost-effective data-driven approach to customer retention, with the ultimate goals of improving customer lifetime value and significantly reducing churn rates. This holistic approach ensures that the project not only meets immediate needs for targeted retention efforts, but also contributes to long-term strategic improvements in customer satisfaction and loyalty.

2. Data Understanding

The project utilizes a comprehensive dataset comprising 34,586 customer records, combined from ten separate datasets (df0-df9) provided by UTS. While the exact data collection methods are unknown, it's assumed that all data was collected simultaneously. The combined dataset contains 31 columns, encompassing a wide range of customer attributes including personal information, account details, viewing habits, and subscription characteristics. Notably, the target variable 'Churn' shows a 17.75% churn rate (6,140 churned customers out of 34,586), indicating a class imbalance.

The data presents some challenges, including missing values in 'Postcode' and 'State' (10.9% each) and 'Ethnicity' (31.9%). These missing values, particularly in location data, appear to be related to military or diplomatic addresses abroad. The dataset includes both categorical (e.g., SubscriptionType, PaymentMethod) and numerical (e.g., MonthlyCharges, ViewingHoursPerWeek) features, providing a rich basis for analysis. However, certain columns (like personal identifiers) pose privacy risks and were excluded during the data preparation stage. The 'Ethnicity' column, despite its potential interest, was also omitted due to high missing values and ethical considerations regarding bias and discrimination. This diverse and comprehensive dataset, despite its limitations, offers valuable insights into customer behavior and characteristics, forming a solid foundation for developing a churn prediction model.

3. Data Preparation

The initial data preparation phase focused on ensuring data privacy and relevance while maintaining the integrity of the dataset for churn prediction. Key steps included the removal of personally identifiable information such as names, addresses, and customer IDs, as well as the exclusion of potentially sensitive demographic data like ethnicity. This approach not only aligns with ethical AI practices but also mitigates risks associated with bias and discrimination. The decision to exclude features with significant missing data, such as postcode and state, was made to maintain data quality and consistency. The dataset was then strategically split into training (60%), validation (20%), and testing (20%) sets, a crucial step to prevent data leakage and ensure unbiased model evaluation.

Feature selection and engineering were conducted with a focus on behavioral and financial factors directly relevant to churn prediction. Mutual Information and Random Forest importance scores were used to identify the most predictive features. Initially, the model relied on raw features encompassing financial metrics, viewing habits, and membership duration to establish a solid baseline. As the project progressed, feature engineering was introduced, exemplified by the creation of the 'ChargesToViewingRatio' feature in Experiment 4, which aimed to capture the relationship between customer spending and engagement. Throughout the process, careful attention was paid to data quality, with checks for missing values, duplicates, and outliers revealing no significant issues requiring immediate attention. For the logistic regression baseline model, numerical features were scaled to ensure fair contribution and optimal model performance, addressing the algorithm's sensitivity to feature scales.

4. Modeling

The approach to churn prediction involved a systematic exploration of various machine learning algorithms, progressively refining the model selection based on performance and business requirements. The modeling process consisted of six key experiments:

Experiment 0 (Baseline) - Logistic Regression:
The process started with a logistic regression as the baseline model due to its simplicity, interpretability, and effectiveness in binary classification tasks. This choice allowed for establishing a performance benchmark and gain initial insights into feature importance. The model's sensitivity to feature scales required standardization of numerical features.

Experiment 1 - Random Forest:
Building on the baseline, I implemented a Random Forest classifier. This algorithm was chosen for its ability to handle non-linear relationships, provide feature importance rankings, and generally perform well with minimal hyperparameter tuning. Random Forest also offers robustness against overfitting, making it suitable for the moderate-sized dataset.

Experiment 2 and 3 – XGBoost, Gradient Boosting and Balanced Random Forest
The random forest did not succeed in predicting churns so I explored XGBoost, an advanced implementation of gradient boosting, with typically high performance on structured data and effective in handling imbalanced datasets. Then Gradient Boosting was chosen for its potential to capture complex patterns in the data while maintaining high accuracy. However, both failed to significantly increase churn prediction. The inclusion of Balanced Random Forest was motivated by the need to address the class imbalance in the churn data more effectively. The model increased the churn recall from under 0,10 in the other models to 0,64, although the accuracy decreased from 0,8+ to only 0,6183.

Experiment 4 - Balanced Random Forest with Feature Engineering:
Based on the promising results of Balanced Random Forest, I focused on optimizing its performance through feature engineering, introducing a new feature, 'ChargesToViewingRatio', and experimented with various feature combinations to enhance the model's predictive power.

Experiment 5 - Balanced Random Forest with Hyperparameter Tuning:
In the final experiment, I performed extensive hyperparameter tuning on the Balanced Random Forest model, utilizing RandomizedSearchCV for broad exploration of the hyperparameter space, followed by a more focused GridSearchCV. This approach allowed for fine-tuning the model for optimal performance.

Throughout these experiments, the model selection process was guided by several key considerations:

- Performance metrics - focused on balancing precision and recall, with particular emphasis on improving recall to identify potential churners effectively.
- Business impact – evaluate models based on their potential to reduce churn while considering the economic viability of the retention strategy.
- Handling class imbalance - prioritizing algorithms and techniques that could deal with the imbalanced dataset.

5. Evaluation

a. Results and Analysis

Experiment 0 (Baseline) - Logistic Regression

```
Accuracy: 0.8261
Precision: 0.5463
Recall: 0.0486
F1 Score: 0.0893
AUC-ROC: 0.7200
```

Classification Report:

	precision	recall	f1-score	support
0	0.83	0.99	0.90	5704
1	0.55	0.05	0.09	1213

Experiment 1 - Random Forest

```
Accuracy: 0.8183
Precision: 0.4052
Recall: 0.0775
F1 Score: 0.1301
AUC-ROC: 0.6647
```

Classification Report:

	precision	recall	f1-score	support
0	0.83	0.98	0.90	5704
1	0.41	0.08	0.13	1213

Experiment 2 – XGBoost

```
Accuracy: 0.8154
Precision: 0.3919
Recall: 0.0956
F1 Score: 0.1537
AUC-ROC: 0.6723
```

Classification Report:

	precision	recall	f1-score	support
0	0.83	0.97	0.90	5704
1	0.39	0.10	0.15	1213

Experiment 3 Gradient Boosting and Balanced Random Forest

```
Gradient Boosting Results:
Accuracy: 0.8269
Precision: 0.5769
Recall: 0.0495
F1 Score: 0.0911
AUC-ROC: 0.7105

Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.99      0.90      5704
           1       0.58      0.05      0.09      1213
```

```
Balanced Random Forest Results:
Accuracy: 0.6183
Precision: 0.2611
Recall: 0.6430
F1 Score: 0.3714
AUC-ROC: 0.6825

Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.61      0.73      5704
           1       0.26      0.64      0.37      1213
```

Experiment 4 - Balanced Random Forest with Feature Engineering

```
Balanced Random Forest Results:
Accuracy: 0.6488
Precision: 0.2842
Recall: 0.6603
F1 Score: 0.3974
AUC-ROC: 0.7107

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.65      0.75      5704
           1       0.28      0.66      0.40      1213
```

Experiment 5 - Balanced Random Forest with Hyperparameter Tuning

```
Balanced Random Forest Results:
Accuracy: 0.6536
Precision: 0.2927
Recall: 0.6884
F1 Score: 0.4107
AUC-ROC: 0.7250

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.65      0.75      5704
           1       0.29      0.69      0.41      1213
```

The baseline logistic regression model showed high accuracy, but very low recall, indicating poor performance in identifying potential churners. Each subsequent model improved on recall, with the Balanced Random Forest models showing significant improvements. The trade-off between precision and recall became evident, with higher recall often accompanied by lower precision. The final tuned Balanced Random Forest model achieved the best balance between precision and recall, as indicated by the highest F1 score and AUC-ROC.

The model's performance on the test set (Accuracy: 0.6539, Precision: 0.2967, Recall: 0.6758, F1 Score: 0.4124, AUC-ROC: 0.7249) is very close to its performance on the validation set, indicating good stability and generalization

b. Business Impact and Benefits

Project Goal Achievement

The project partially met its goals. We successfully developed a machine learning model capable of predicting customer churn with improved accuracy, achieving a recall of 0.6758 on the test set, which indicates a strong ability to identify potential churners. However, the model falls short of making the discount strategy immediately economically viable over a three-month period. While the model identifies a significant portion of likely churners, the high number of false positives leads to a short-term financial loss. The goal of reducing the overall churn rate by at least 50% through targeted interventions is theoretically achievable given the model's recall, but the actual impact would depend on the effectiveness of the retention strategies applied to identified at-risk customers. The third goal of implementing a cost-effective retention strategy with a net positive financial impact was not met in the short term, as the model currently results in a net loss over the initial three-month period.

Financial Impact and Break-Even Analysis

The final model, when applied to the test dataset, shows that implementing the discount strategy would result in a revenue of $191,231.25 compared to $212,812.50 if no action is taken, leading to a short-term loss of $21,581.25 over three months. However, this analysis is based solely on a three-month timeframe and doesn't account for potential long-term benefits. To calculate the break-even point, let's consider the average monthly charge of $12.5. The three-month 50% discount costs $18.75 per customer (3 * $12.5 * 0.5). For the strategy to break even, a retained customer would need to stay for an additional 1.5 months at full price after the discount period (18.75 / 12.5 = 1.5). Therefore, if a churner stays for a total of 4.5 months (3 months discounted + 1.5 months full price), the strategy becomes financially beneficial. Any retention beyond 4.5 months would contribute to net positive revenue.

c. Data Privacy and Ethical Concerns

In developing the churn prediction model, I prioritized ethical considerations to ensure fair and responsible use of customer data. A key decision was the exclusion of sensitive demographic information, particularly ethnicity data, to prevent potential discrimination and bias in the model's predictions. This approach aligns with principles of fairness in AI, ensuring that the churn predictions and subsequent retention strategies are based solely on customer behavior and engagement metrics rather than personal characteristics (Mehrabi et al., 2021). By focusing on behavioral data, I aim to avoid potential biases that could arise from the use of demographic information. Furthermore, transparency is emphasized in the modeling process, particularly through the use of interpretable models in earlier stages and consistent feature importance analysis. This transparency allows for a clear understanding of the factors influencing churn predictions, enabling scrutiny and validation of the model's decision-making process.

However, I acknowledge that ethical challenges persist even with these precautions. There's a potential risk of inadvertently favoring certain customer groups in the retention efforts, which could lead to unequal treatment. Additionally, over-reliance on model predictions at the expense of other customer retention strategies poses a risk of overlooking important qualitative factors in customer relationships. To mitigate these risks, I recommend implementing ongoing monitoring of model performance across different customer segments, conducting regular ethical audits, and maintaining strict data handling protocols. It is also crucial to ensure that the data usage complies with customer agreements and relevant data protection regulations, maintaining a balance between leveraging data for business insights and respecting customer privacy.

6. Conclusion

Key Findings and Insights
This churn prediction project has yielded valuable insights into customer behavior and the potential of machine learning in customer retention strategies. The final Balanced Random Forest model demonstrated a strong ability to identify potential churners, achieving a recall of 0.6758 on the test set. This means the model can correctly identify about 68% of customers who are likely to churn, a significant improvement over baseline methods. The model consistently highlighted AccountAge and AverageViewingDuration as the most crucial factors in predicting churn, providing actionable insights for customer retention strategies.

Project Success and Stakeholder Requirements:
The project partially met its initial goals. We successfully developed a model capable of identifying a significant portion of potential churners, which addresses the marketing team's need for targeted retention efforts. However, the economic viability of the proposed discount strategy in the short term was not achieved due to the high number of false positives. The finance department's requirement for a cost-effective solution is not immediately met, as the model currently results in a short-term net loss. Nevertheless, the break-even analysis suggests that the strategy could become profitable if retained customers stay for just 1.5 months beyond the discount period, aligning with the long-term value perspective of customer retention.

Future Work and Recommendations

1. Precision Improvement: Future iterations should focus on improving the model's precision without significantly sacrificing recall. This could involve more advanced feature engineering, ensemble methods, or exploring deep learning approaches.
2. Segmented Retention Strategies: Develop varied retention strategies based on the predicted probability of churn, potentially offering different incentives to different risk segments.
3. Long-term Impact Study: Implement a pilot program to track the long-term retention rates of customers identified by the model and given discounts. This will provide real-world data on the strategy's effectiveness beyond the initial three-month period.
4. Incorporate Time-Series Data: If possible, integrate historical data to capture trends in customer behavior over time, potentially improving the model's predictive power.
5. Ethical Considerations: Continue to prioritize fairness and transparency in the model, regularly auditing for any unintended biases and ensuring compliance with data protection regulations.

In conclusion, while the project has not fully achieved its economic goals in the short term, it has laid a strong foundation for data-driven customer retention strategies. The insights gained and the predictive capability developed provide a valuable tool for targeted interventions. With further refinement and a focus on long-term customer value, this approach has the potential to significantly impact customer retention and overall business performance. The next steps should involve a careful balance of model improvement, strategy refinement, and real-world testing to fully realize the potential of this churn prediction model.

7. References

Canal, Alexandra. "Americans Are Canceling More Streaming Plans as Prices Balloon." *Yahoo Finance*, 23 Aug. 2023, finance.yahoo.com/news/americans-are-canceling-more-streaming-plans-as-prices-balloon-153035743.html?ref=churnkey.co&guccounter=1.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys*, *54*(6), 1–35. https://doi.org/10.1145/3457607